2.2 Methods	Experiments 2.2.1 Designing and conducting experiments, including field and laboratory experiments. 2.2.2 Independent and dependent variables. 2.2.3 Experimental and null hypotheses. 2.2.4 Directional (one-tailed) and non-directional (two-tailed) tests and hypotheses. 2.2.5 Experimental and research designs: repeated measures, independent groups and matched pairs. 2.2.6 Operationalisation of variables, extraneous variables and confounding variables. 2.2.7 Counterbalancing, randomisation and order effects. 2.2.8 Situational and participant variables. 2.2.9 Objectivity, reliability and validity (internal, predictive and ecological). 2.2.10 Experimenter effects, demand characteristics and control issues.
	2.2.11 Quantitative data analysis • Analysis of quantitative data: calculate measures of central tendency, frequency tables, measures of dispersion (range and standard deviation), percentages. • Graphical presentation of data (bar graph, histogram).
	2.2.12 Decision making and interpretation of inferential statistics • Non-parametric test of difference: Mann-Whitney U and Wilcoxon. • Probability and levels of significance ($p \leq .10$ $p \leq .05$ $p \leq .01$). • Observed and critical values, **use of critical value tables** and sense checking of data. • One- or two-tailed regarding inferential testing. • Type I and type II errors. • Normal and skewed distribution.
	2.2.13 Case study of brain-damaged patients, including Henry Molaison (HM) and the use of qualitative data, including strengths and weaknesses of the case study.